

Multi-View Spatial Attention Embedding for Vehicle Re-Identification

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Abstract—Vehicle Re-Identification (Re-ID) is a challenging vision task mainly because the appearance of a vehicle varies dramatically under different viewpoints. Moreover, different vehicles with the same model and color commonly show similar appearance, thus are hard to be distinguished. To alleviate negative effects of viewpoint variance, we design a multi-view branch network where each branch learns a viewpoint-specific feature without parameter sharing. Being able to focus on a limited range of viewpoints, this viewpoint-specific feature performs substantially better than the general feature learned by a uniform network. To further differentiate visually similar vehicles, we strengthen the discriminative power on their subtle local differences by introducing a spatial attention model into each feature learning branch. The multi-view feature learning and spatial attention learning compose our neural network architecture, which is trained end to end with the softmax loss and triplet loss, respectively. We evaluate our methods on two large vehicle Re-ID datasets, *i.e.*, *VehicleID* and *VeRi-776*, respectively. Extensive experiments show that our methods achieve promising performance. For example, we achieve mAP accuracy of 76.78% and 72.53% on *VehicleID* and *VeRi-776* dataset respectively, substantially better than current state-of-the-art.

Index Terms—Vehicle re-identification, convolutional neural network, attention, multi-view.

I. INTRODUCTION

VEHICLE Re-Identification (Re-ID) targets to match a query vehicle image against a large vehicle image gallery set. It is commonly regarded as an important technique to achieve efficient traffic control and smart city surveillance, especially for the scenario, where licence plates are not available or reliable. Compared with other fine-grained recognition

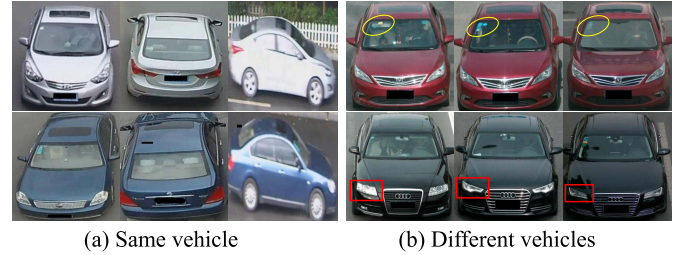


Fig. 1. Illustration of two challenges faced by vehicle Re-ID task: (a) viewpoint variance leads to dramatically different appearances of the same vehicle, and (b) lots of different vehicles are visually similar to each other.

tasks in surveillance video like person re-identification and face recognition [1]–[11], vehicle Re-ID is still a relatively under-explored task.

The challenges of vehicle Re-ID are mainly due to two facts: a) the same vehicle would show substantially different appearances under different viewpoints, and b) different vehicles with the same model and color may have inconspicuous differences under the same viewpoint. We illustrate the effect of viewpoint variation in Fig. 1 (a), where the appearance of a vehicle varies dramatically under different viewpoints. Fig 1 (b) shows some vehicles of the same model in each row. It is clear that, those vehicles exhibit similar appearance and can only be distinguished by local details on windshields or headlights.

Some of vehicle Re-ID approaches [12]–[15] adopt distance metric learning loss or classification loss in a brute-force way to train a global feature. Although these works have achieved remarkable success, they rely on global cues, which are not discriminative enough to differentiate visually similar vehicles. To handle this problem, some other works [16]–[20] learn local parts or apply attention to highlight discriminative local details. For example, Liu *et al.* [17] proposed a Region-Aware deep Model to jointly learn deep features from both global and local regions. Teng *et al.* [18] designed an attention module to refine feature maps learned by CNN. Wang *et al.* [16] pre-trained a region proposal module to produce response maps of 20 vehicle keypoints, which are hence used to extract local features to enhance the discriminative power. Zhou and Shao [19] proposed a viewpoint-aware attention model and focused on specific areas from different viewpoints. This algorithm boosts the robustness of visual feature to viewpoint

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variances. However, it is complicated and difficult to train. More detailed review of existing works will be presented in Sec. II.

This paper targets to address the challenges illustrated in Fig. 1 with a more efficient CNN design. As different viewpoints dramatically change the appearance of a vehicle, it is not appropriate to use a uniform network for feature extraction under different viewpoints. In other words, it is difficult to map different viewpoints of one vehicle into close feature vectors with the same network. We hence design a multi-branch feature extraction network, where each branch learns features for a limited range of viewpoints. Different branches are trained with different parameters to boost the robustness to viewpoint changes. Features of different branches are finally fused to obtain the final feature for vehicle Re-ID. Compared with existing uniform feature extraction network, this design encourages the network to focus on learning discriminative viewpoint-invariant features.

Explicitly extracting part features either requires extra key-point annotations or labeled bounding boxes. To alleviate the expensive data annotation and differentiate visually similar vehicles, we embed spatial attention modules to learn discriminative features. Spatial attention modules compute both local and global attentions. Discriminative local features are learned by identifying local regions with the parameter-free local attention module. A global attention module is further applied to filter cluttered backgrounds, which also boosts the vehicle Re-ID performance.

The multi-view feature learning and spatial attention learning compose our neural network architecture, which is trained with softmax loss and triplet loss, respectively. We tested our approach on two widely-used vehicle Re-ID datasets, *i.e.*, *VehicleID* [12] and *VeRi-776* [21], respectively. Our method achieves competitive performance in comparison with recent works. For example, it achieves rank-1 accuracy of 72.58% on the *VehicleID* dataset, significantly outperforming the recent VAMI [19] by 25%, which also considers multi-view feature embedding. It also performs better than the OIFE [16], which uses extra local keypoint annotations and spatial-temporal cues for feature extraction. We hence could summarize our contributions as follows:

- 1) We propose a multi-view branch network, where each branch focuses on a limited range of viewpoint changes to facilitate the learning of discriminative features for visually similar vehicles. With the multi-branch network structure, the resulting feature presents strong discriminative power and robustness to viewpoint variations.
- 2) Global-local spatial attention learning is proposed to alleviate cluttered backgrounds and discover subtle local differences among different vehicles. Compared with existing local feature learning strategies in vehicle Re-ID works, our method performs well without requiring extra bounding box annotations.
- 3) Extensive comparisons with recent works on two large vehicle Re-ID datasets demonstrate the effectiveness of proposed methods.

The remainder of this paper is organized as follows. Sec. II reviews related works. Sec. III presents the detailed

descriptions of our algorithms. Sec. IV introduces our experiments, followed by the conclusions in Sec. V.

II. RELATED WORK

According to their adopted deep models, current vehicle Re-ID works can be summarized into three categories, *i.e.*, 1) global feature learning, 2) regional feature learning, and 3) extra spatial-temporal cues utilization, respectively. The following parts review those three categories of works.

A. Global Feature Learning

As Re-ID tasks can be tackled by measuring the feature distance, distance metric learning is commonly adopted to learn discriminative global features. Several variants of metric loss have been widely applied in related tasks, including contrastive loss [22], triplet loss [23]–[25], and quadruplet loss [26]. Center loss [27] is another metric loss which enforces the inter-class dispersion and intra-class compactness.

Re-ID models can also be trained with classification tasks, where each ID is treated as a category. Classification loss like softmax loss is also commonly used to train visual features. Large margin softmax loss [10], A-Softmax loss [28] and large margin cosine loss [11] are three modified versions of softmax loss. The research of Yan, *et al.* shows that, combining metric loss and classification loss achieves better Re-ID performance [13].

For vehicle Re-ID task, Liu *et al.* [12] proposed a Coupled Clusters Loss (CCL) to enhance the discriminative power of learned deep features. Yang *et al.* [29] learned rich semantic features of vehicles from car model classification and car model verification tasks. Liu *et al.* [21] fused attributes and color features of vehicles and integrate visual appearance, plate and spatiotemporal information for vehicle Re-ID. Bai *et al.* [15] proposed a group sensitive triplet embedding approach to model the inter-class dissimilarity as well as the intra-class invariance during neural network training. Guo *et al.* [30] proposed a coarse-to-fine ranking loss to learn a structured deep feature embedding for vehicle Re-ID. Kumar *et al.* [31] introduced a triplet sampling strategy to boost the performance of triplet loss. All those metric learning and classification algorithms mentioned above optimize the feature distance between global features and do not consider the local details.

This paper tackles vehicle Re-ID with the multi-branch CNN and spatial attention learning. We adopt the commonly used triplet loss and softmax loss as our training loss to show the validity of proposed methods. Nevertheless, our framework can be combined with other recent metric learning approaches to explore better Re-ID performance.

B. Regional Feature Learning

Local details are important for differentiating visually similar identities. Another closely related Re-ID task, *i.e.*, person Re-ID commonly uses part cues to extract discriminative visual features. In particular, human pose estimation and landmark detection have achieved impressive progress [3].

Several recent works fuse global feature and regional features extracted from body parts for person Re-ID [32]. Yao *et al.* [33] clustered coordinates of maximum activations on feature maps to locate regions of interest. Zhao *et al.* [34] proposed the human part-aligned representation to handle the body parts misalignment issue. Liu *et al.* [17] also showed that part model is valid for vehicle Re-ID. He *et al.* [35] proposed a part-regularized discriminative feature preserving method to enhance the discriminative power to subtle local differences. However, it needs additional bounding box detection network for part localization, which is not applicable for side view vehicle images.

Recently, regional attention learning has been adopted in many vision tasks to handle the misalignment issues and filter noises [36]–[42]. Zhu *et al.* [38] learned a spatial attention mask to improve the feature representation and the tracking accuracy. Chen *et al.* [36] introduced a spatial attention model to weight the multi-scale features at each pixel location. The attention model learns to weight objects of different scales. Wang *et al.* [39] proposed a Residual-Attention-Network which stacks multiple attention modules to obtain better image classification accuracy. Li *et al.* [40] proposed the idea of jointly learning multi-granularity attention selection and feature representation for optimising person re-id in deep learning.

Some vehicle Re-ID works also introduce attention modules. Wang *et al.* [16] developed a framework utilizing key-point annotations on vehicles to learn viewpoint invariant features. Their model combines *VeRi-776* [21], *VehicleID* [12], *CompCars* [29] and *Cars21k* [43] into one large training set, which yields promising performance. Teng *et al.* [18] designed an attention module to refine the feature maps in CNN. Zhou and Shao *et al.* [19] proposed a viewpoint-aware attention model and focus on specific areas from different viewpoints. All those regional attention methods either design a simple spatial attention mask to weight the original feature map or stack multiple attention modules to enrich the final feature representation.

Our spatial attention module shares certain similarity with attention methods existing vehicle Re-ID works. However, our method differs with those works in following aspects. 1) We consider both local and global attention cues, which learn complementary features in a simple way. 2) Spatial attention learning is embedded in the multi-branch architecture to learn specific local cues for different viewpoints. Those two unique properties help us to obtain more discriminative feature representations.

C. Spatial Temporal Cues Utilization

It could be difficult to conduct vehicle Re-ID only with appearance cues in some scenarios. Liu *et al.* [21], [44] proposed a vehicle Re-ID dataset, *i.e.*, the *VeRi-776*, which contains extra spatial-temporal cues. Another work [21] uses licence plate images and spatial-temporal cues to learn similarities between vehicle images. Shen *et al.* [14] also used spatio-temporal cues for improving the Re-ID accuracy. A chain MRF [45] model is used to generate a

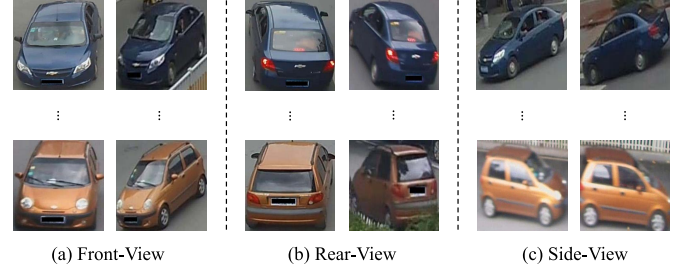


Fig. 2. Illustration of three typical viewpoints of vehicle images. (a) Shows images with front and front-side view. (b) Shows images with rear and rear-side view. (c) Shows image with the side view. Our work hence defines three categories for viewpoints.

visual-spatial-temporal path. The confidence score of each path is then estimated by LSTM [46]. Some other researchers, *e.g.*, Wang *et al.* [16], Zhou and Shao [19], and Liu *et al.* [47] used spatial-temporal constraints to refine retrieval results, respectively. Although spatial and temporal cues could be helpful for vehicle Re-ID, they are not always available, which limits the application of those algorithms. In this work, we focus on learning visual features, and do not consider the utilization of spatial-temporal cues.

III. METHODOLOGY

A. Formulation

Given a training set $X = \{x_i, y_i\}_{i=1}^N$ containing N vehicle images, where each image x_i is labeled with an identification label y_i . Each vehicle image may present distinct viewpoints, *e.g.*, front-view, side-view, and rear-view, as illustrated in Fig. 2. We hence learn viewpoint-invariant feature to make images with different viewpoints comparable in the Re-ID procedure. The basic idea is to treat different viewpoints with different feature extractors, which are trained to produce similar features for the same vehicle, and distinct features for different vehicles.

We design a multi-view branch network for vehicle feature extraction. Suppose there are V typical viewpoints for vehicle images, we build a network with V branches, each of which is designed for the feature extraction under a specific viewpoint. Given an input image x_i , we denote the feature map extracted by one branch as, *i.e.*,

$$F_{v,i} = \mathbb{H}_v(x_i), \quad v \in [0, V], \quad (1)$$

where v is the index of a branch, $\mathbb{H}_v(\cdot)$ denotes the feature extractor for the v -th branch, and $F_{i,v}$ is a tensor with size $h \times w \times c$. The Global Average Pooling (GAP) layer can be applied on $F_{i,v}$ to produce the feature vector for a specific viewpoint,

$$f_{v,i} = \text{GAP}(F_{v,i}), \quad (2)$$

where $f_{v,i} \in \mathbb{R}^c$ is the feature vector for viewpoint v .

As vehicle images may present different viewpoints, we could represent the viewpoint of one vehicle image with a probabilistic representation $\{p_v\}_{v=1}^V$, $p_v \in [0, 1]$, where V

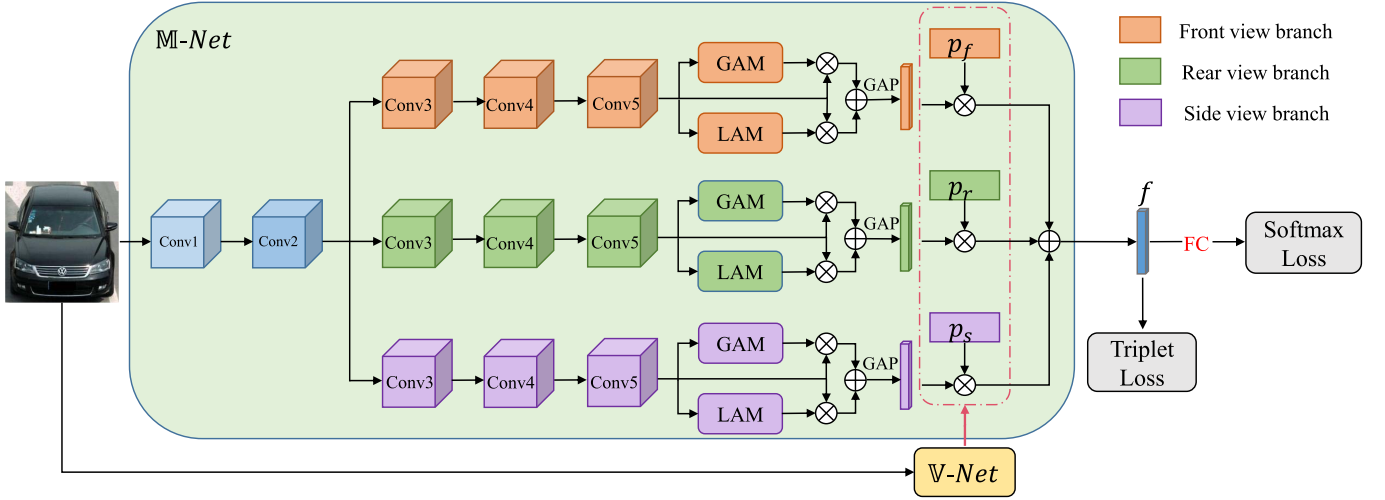


Fig. 3. Illustration of the proposed Multi-View Attention Network (MVAN) for vehicle Re-ID. $\mathbb{V}$ -Net estimates the viewpoints of input vehicle images. $\mathbb{M}$ -Net enforces each branch to focus on feature learning for a specific viewpoint to learn viewpoint-invariant feature. Global Attention Module (GAM) and Local Attention Module (LAM) are applied to refine the original features. $\otimes$ denotes element-wise product. $\oplus$ denotes element-wise sum.

denotes the total number of viewpoint categories. The final fused vehicle feature could be represent as,

$$f_i = \sum_{v=1:V} p_v \cdot f_{v,i}, \quad (3)$$

where the probabilistic vector could be predicted by a viewpoint estimation module, i.e.,

$$\{p_v\}_{v=1}^V = \mathbb{V}(x_i), \quad (4)$$

where $\mathbb{V}(\cdot)$ denotes viewpoint estimation module with the original image as input.

Discussions: The above equations enforce each branch to focus on feature learning for a specific viewpoint, thus are beneficial to boost the robustness to viewpoint variances and discriminative power for visually similar vehicles. For instance, for the front-view images, the branch corresponding to front-view will dominate the feature extraction. For side-view images, the branch of side-view will dominant the feature extraction. The network will be trained to ensure 1) images of the same vehicle from different viewpoints would have similar features, 2) images of different vehicles under the same viewpoint would have distinct features. This method differs with previous works that commonly use a uniform extractor for feature extraction under different viewpoints.

To differentiate visually similar vehicles, we further embed spatial attention modules to learn discriminative features. Spatial attention modules compute local and global attention maps, which are used to identify discriminative local regions and filter noises like cluttered backgrounds, respectively. The global and local attention maps are defined as $h \times w \times 1$ sized matrix. They are posted on the feature map of each branch, i.e.,

$$\hat{F}_{v,i} = F_{v,i} \odot (G_{v,i} + L_{v,i}), \quad (5)$$

$$G_{v,i} = \text{GAM}_v(F_{v,i}), \quad (6)$$

$$L_{v,i} = \text{LAM}_v(F_{v,i}), \quad (7)$$

where $G_{v,i}$ and $L_{v,i}$ are global and local attention maps on the v -th branch, respectively. $\text{GAM}_v(\cdot)$ and $\text{LAM}_v(\cdot)$ are modules for estimating the global and local attention maps, respectively.

With attention learning, the fused feature in Eq. (3) could be updated as,

$$f_i = \sum_{v=1:V} p_v \cdot \hat{f}_{v,i}. \quad (8)$$

$$\hat{f}_{v,i} = \text{GAP}(\hat{F}_{v,i}). \quad (9)$$

We use softmax loss and triplet loss computed based on f to train the neural network, i.e., the feature extractor $\mathbb{H}$, attention models GAM and LAM, respectively. The training loss could be represented as

$$\mathcal{L}(f) = \mathcal{L}_{id}(f) + \alpha \cdot \mathcal{L}_{triple}(f), \quad (10)$$

where α weights those two loss functions. $\mathcal{L}_{id}(\cdot)$ denotes the softmax loss computed with vehicle ID labels. $\mathcal{L}_{triple}(\cdot)$ denotes the triplet loss.

As shown in our experiments, combining two loss functions brings performance gains. More details of our implementation to $\mathbb{H}$, GAM, LAM, as well as the viewpoint estimator $\mathbb{V}$ are presented in the following section.

B. Implementation

The above formulation leads to the Multi-View Attention Network (MVAN) illustrated in Fig. 3, where the MVAN consists of the multi-view branch network, spatial attention block, and a viewpoint estimation module, respectively. The following parts introduce the implementation of those components.

1) Viewpoint Estimation: To implement the viewpoint estimation, we first define the viewpoint categories, then pre-train a CNN as the viewpoint estimator. Commonly used vehicle Re-ID datasets like *VehicleID* [12] and *VeRi-776* [21] do not provide viewpoint annotations. As the number of viewpoint categories, i.e., the V , largely decides the complexity of our model, we tend to set V a small value. As shown in Fig. 2, vehicle images commonly show three typical viewpoints, we hence set $V = 3$. This differs from existing works [19], [48], which define 5 and 8 viewpoints, respectively.

We manually annotate some viewpoint labels to train the viewpoint estimator. *Veri-776* dataset contains 37,778 vehicle images in the training data. To simplify the annotation, only 2,576 training images on *Veri-776* are annotated with three labels, *i.e.*, front, rear and side. A three-class viewpoint estimator is hence trained on *Veri-776*. A two-class estimator is also trained using front and back-view labels on *Veri-776*, which is hence used to generate the two-viewpoint labels on *VehicleID*. Therefore, the manual annotation is only conducted on *Veri-776*, and could be finished with limited costs.

As shown in Fig. 3, we use the $\mathbb{V}$ -Net as the viewpoint estimator. Because viewpoint recognition is not a challenging task, we implement $\mathbb{V}$ -Net based on VGG_CNN_M_1024 [49]. We keep the original convolutional layers in VGG_CNN_M_1024 and simplify two fully-connected (fc) layers. Specifically, we apply GAP layer after the fifth convolutional layer, which is followed by a fc layer to produce a 512-dim feature. Finally, a new fc layer with three outputs corresponding to three viewpoints are computed on the 512-dim feature. Compared to the original VGG_CNN_M_1024, our network presents better compactness.

The $\mathbb{V}$ -Net is trained off-line and is not updated during feature learning. As shown in Fig. 3, $\mathbb{V}$ -Net predicts three probability scores p_f , p_r and p_s for the front-view, rear-view, and side-view, respectively.

2) *Multi-View Branch Network*: The multi-view branch network, which is named as the $\mathbb{M}$ -Net is illustrated in Fig. 3. We choose VGG16 [50] as the backbone. The three branches share the first several convolutional blocks, then have their own feature extraction branches. Each feature extraction branch composes of several convolutional blocks and a spatial attention block. As shown in the figure, each branch produces a feature vector with a GAP layer. The three feature vectors are fused with weights learned by the $\mathbb{V}$ -Net. Two fully-connected layers are applied on the fused layer to predict the vehicle ID labels.

As shown in Eq. (10), we use softmax loss and triplet loss to train the $\mathbb{M}$ -Net. We compute the softmax loss as:

$$\mathcal{L}_{id}(f_i) = \sum_{j=1}^t -y_i(j) \cdot \log\left(\frac{\exp(\omega_j^T f_i)}{\sum_{k=1}^t \exp(\omega_k^T f_i)}\right), \quad (11)$$

where f_i and y_i refer to the extracted feature vector and vehicle ID label for the i -th training image. t denotes the number of vehicle categories in the training set, and ω_k denotes the classifier parameters of the k -th category. Note that, the groundtruth label y_i is an one-hot vector.

Triplet loss enforces one feature to be closer to another feature from the same vehicle, than to a feature from any other vehicles. The triplet loss can be defined as:

$$\begin{aligned} \mathcal{L}_{triplet}(f^{(a)}, f^{(p)}, f^{(n)}) \\ = \max(0, d(f^{(a)}, f^{(p)}) - d(f^{(a)}, f^{(n)}) + m), \end{aligned} \quad (12)$$

where $d(\cdot)$ represents the L_2 distance, superscripts a , p , and n denote the anchor sample, positive sample and negative sample, respectively. m is a constant threshold value.

Features learned by each branch are weighted by predicted scores of $\mathbb{V}$ -Net. This design chooses the proper

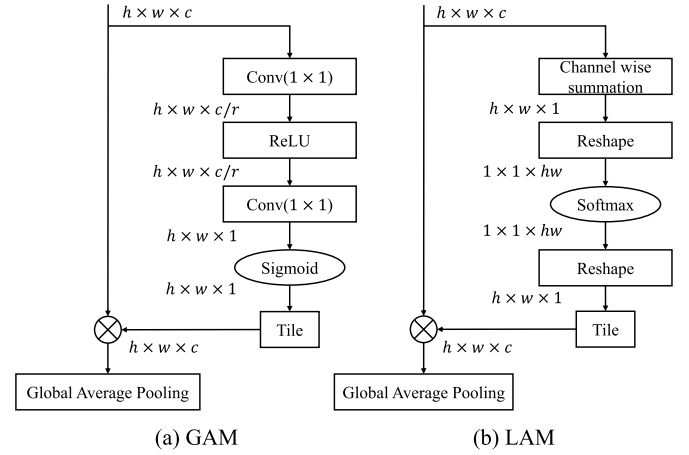


Fig. 4. The structure of spatial attention model. (a) Global Attention Module (GAM). (b) Local Attention Module (LAM).

feature extraction branch, *e.g.*, the “front-view” branch will be selected for vehicles with front-view. This design also modulates the gradient back-propagation during the network training, which allows for learning viewpoint specific feature extractors. For instance, for a training sample with a strong “front-view” prediction, the “front-view” branch would receive strong gradient updates.

3) *Spatial Attention Block*: As vehicles with the same model and color commonly show similar appearances, we further design spatial attention blocks to identify discriminative local cues. Besides identifying local details, we also apply spatial attention to filter noises in feature maps.

In each branch of the $\mathbb{M}$ -Net, two attention blocks are designed to focus on global and local cues respectively. The local attention module identifies discriminative local cues. The global attention module identifies informative regions and filters cluttered backgrounds. As shown in Fig. 3, local and global attention blocks are inserted after the final convolutional block in each branch. The structures of those two attention modules are shown in Fig. 4.

a) *Global attention module*: (GAM) is designed to make the network pay attention to vehicle foregrounds and filter backgrounds. We design two convolution layers to learn a global vehicle mask. The ReLU layer and Sigmoid layer are applied after the first and second convolution layer respectively. The structure of GAM is illustrated in Fig. 4 (a).

For the v -th branch, the input of GAM is a 3D tensor with size $F_v \in R^{h \times w \times c}$, where h , w and c denote the size of height, width and channel of feature maps, respectively. Based on F_v , we apply two 1×1 convolution layers with c' and 1 channel outputs to obtain an attention mask $M_v \in R^{h \times w \times 1}$. The value at the spatial location $(i, j, 1)$ of mask M can be computed as,

$$M_v(i, j, 1) = \sum_{p=1}^{c'} w_v^{(2)}(p) \cdot F'_v(i, j, p), \quad (13)$$

$$F'_v(i, j, p) = \sum_{k=1}^c w_v^{(1)}(k) \cdot F_v(i, j, k), \quad (14)$$

where $F_v(i, j, k)$ is the value of feature map at the location (i, j) of k -th channel, F'_v is the feature map with c' channels.

$F_v(i, j, k) \geq 0$. $w_v^{(1)}$ and $w_v^{(2)}$ are the 1×1 convolution kernel parameters. In (13) $c' = c/r$, where r is experimentally set to 16.

After computing the mask M_v we apply Sigmoid function to normalize the value of M_v between 0 and 1. The updated attention map G_v can be computed as,

$$G_v(i, j, 1) = \frac{1}{1 + e^{-M_v(i, j, 1)}}, \quad (15)$$

where the matrix G is used as the final global attention mask.

b) Local attention module: (LAM) is applied to automatically discover discriminative local regions on vehicles. Fig. 4 (b) shows details of the LAM. As shown in the figure, the input of LAM for the v -th branch is also a 3D tensor $F_v \in R^{h \times w \times c}$. As discussed in previous works [51], CNN training tends to spot and focus on discriminative cues on images. In other words, discriminative regions would correspond to large activations on feature maps. To identify discriminative regions, LAM first computes the summarization over all channels to compute a matrix S , *i.e.*,

$$S_v(i, j, 1) = \sum_{k=1}^c F_v(i, j, k). \quad (16)$$

where the input of the equation, *i.e.*, $F_v(i, j, k)$, is an output from a ReLU layer, making it contain nonnegative values.

To further emphasize the regions with high activations, Softmax function is then applied on the S to compute the final local attention map L_v , *i.e.*,

$$L_v(i, j, 1) = \frac{e^{S_v(i, j, 1)}}{\sum_{i, j} e^{S_v(i, j, 1)}}. \quad (17)$$

With Eq. (15) and Eq. (17), we could compute the global and local attention maps for the v -th branch, which are hence fused to update the original feature map F_v with Eq. (5).

Discussion: Instead of learning a rigid spatial decomposition of input images as in [52], GAM and LAM automatically identify salient regions with arbitrary shape in each vehicle image. This property of GAM and LAM helps to find more useful cues. GAM eliminates the background noises and LAM intensifies the discriminative local details. As shown in Fig. 4 (b), the LAM is computed by channel-wise summation, softmax, and two reshape layers. Therefore, the LAM does not involve extra parameters to be learned. Although LAM is a parameter-free spatial attention layer, it effectively leverages the discriminative cues captured by original CNN.

Fig. 5 shows some examples of global and local attention maps learned by GAM and LAM, respectively. We can observe that, the global attention maps generally cover the vehicle foreground, and have weak responses on backgrounds. The local attention maps pay more attention to some distinctive local regions such as head lights, vehicle logos, windcreens, *etc.* With the combination of global and local attention maps, we obtain more discriminative feature representations. More extensive experiments on the validity of GAM and LAM will be presented in the experimental section.

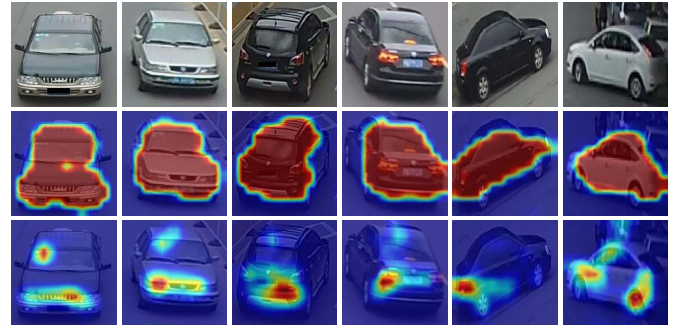


Fig. 5. Visualization of attention maps generated by GAM and LAM trained on *VeRi-776* dataset. The first row shows original images. The second and third rows show corresponding attention maps generated by GAM and LAM, respectively.

C. Training and Inference

The above parts present the implementation of important components in MVAN. The training of MVAN consists of two stages. In the first stage, the $\mathbb{V}$ -Net for vehicle viewpoint classification is trained using Softmax loss with viewpoint labels. In the following stage, we fix $\mathbb{V}$ -Net and train the $\mathbb{M}$ -Net by softmax loss and triplet loss. During inference, given a test probe vehicle image and a set of gallery vehicle images, we first compute feature vector f by forward-feeding each image to the trained MVAN. Then, we compute the distance between query and each gallery image with the L_2 distance, and rank all gallery images according to their distance to the probe image. The top-ranked images are supposed as the true positives of the Re-ID result.

IV. EXPERIMENTS

A. Datasets

We use the following two datasets to verify the performance of proposed methods.

VeRi-776 [21] is collected from real-world surveillance scenarios. It contains about 51,035 bounding boxes of 776 vehicles. *VeRi-776* is split into 576 vehicles for training and 200 vehicles for testing. The standard probe and gallery sets consist of 1,678 and 11,579 images, respectively. Vehicles in this dataset present three viewpoints, *i.e.*, front, rear, and side views, respectively, making it one of the most challenging Re-ID datasets. *VeRi-776* provides annotations of the ID, model, color of vehicles, inter-camera relations, as well as the trajectory information.

VehicleID [12] is also collected by surveillance cameras. It contains 26,267 vehicles and 221,763 images. *VehicleID* provides a large training set with 100,182 images of 13,164 vehicles and a test set with 19,777 images of 2,400 vehicles. Vehicle images in *VehicleID* are either captured from the front-view or the back-view. It also provides the annotations of 250 vehicle models.

B. Implementation Details

Experiments are conducted based on Caffe [53]. We use VGG16 [50] and ResNet50 [54] as backbones, which are

pre-trained on ImageNet [55]. We modify the VGG16 by first removing the original two 4096D fc layers and 1000D classification layer, then adding a GAP layer and a 512D fully-connected feature layer, followed by Batch Normalization and ReLU [56], [57]. A new classification layer is added for vehicle classification. For the ResNet50 backbone, we removed the last spatial down-sampling operation to increase the spatial resolution of the feature map [9], [57]. This helps to preserve more spatial details.

Each image is first resized into 256×256 , then is randomly cropped into a 224×224 sized image. The final features f is used to calculate the triplet loss. The margin m of triplet loss is experimentally set to 0.5. The mean Average Precision (mAP), Top-1 accuracy and Top-5 accuracy are chosen as the evaluation metric. We adopted the SGD optimizer and set the initial learning rate to 0.001 for VGG16 and 0.01 for ResNet50. The learning rate decay factor is 0.2 for every 80,000 iterations. The weight decay factor is set to 0.0002 for VGG16 and 0.0004 for ResNet50. The momentum is set to 0.9. We mainly use the vehicle identification loss and triplet loss for CNN training. The weights for those two losses are set as equal, *i.e.*, $\alpha = 1$ in Eq. (10). We set batch size to 16 and train the model for 400,000 iterations.

We follow existing works [12], [21] to setup the experiments. During testing on *VehicleID*, one image is randomly selected from each vehicle to generate a gallery set with 2,400 images. The remaining 17,377 images are used as query images. The random selection process was repeated for 10 times to obtain the final performance. For the *VeRi-776*, we select 11,579 images as gallery set and use the remaining 1,678 images as query images. We follow the selection strategy in a previous work [21].

C. Ablation Studies

1) *Effects of Multi-View Branch*: To verify the effectiveness of the multi-view feature learning strategy, we first conduct experiments to compare the following algorithms:

- Baseline-1 generates feature representations from VGG16 backbone. No multi-view cues or multi-branch structure is considered for feature learning.
- Baseline-2 has three parallel branches trained independently. Each branch has the same network structure as the Baseline-1. Different branches do not share parameters and their learned features are concatenated as the final feature.
- Multi-View denotes our proposed architecture without the spatial attention block. As shown in Fig. 3, it is implemented based on Baseline-1 by inserting multiple branches after the conv4 stage.
- Multi-View-Binary is the same as Multi-View, but uses one hot vector as the viewpoint weighting parameter. The one-hot vector is generated based on the probabilistic scores computed by viewpoint estimator.

Table I summarizes experimental results. As shown in the table, Baseline-2 performs better than Baseline-1 by concatenating multiple CNNs features. It is also clear that, both Multi-View and Multi-View-Binary achieve further improvements over the Baseline-2. Note that, Baseline-2 involves more

TABLE I
COMPARISON BETWEEN OUR MULTI-VIEW BRANCH ARCHITECTURE AND TWO BASELINES ON *VehicleID* AND *VeRi-776*. IT IS CLEAR THAT, OUR MULTI-VIEW BRANCH FRAMEWORK ACHIEVES THE BEST PERFORMANCE

Dataset	VehicleID		VeRi-776		
Methods	r=1	r=5	r=1	r=5	mAP
Baseline-1	60.63	72.67	76.82	86.71	39.91
Baseline-2	64.81	75.84	80.82	89.57	49.68
Multi-View	66.06	76.13	82.64	90.87	52.26
Multi-View-Binary	65.99	76.15	82.13	90.53	52.29

TABLE II
PERFORMANCE OF BASELINE MODELS TRAINED WITH IMAGES FROM ONE OR TWO VIEWPOINTS

Dataset	VeRi-776		
Methods	r=1	r=5	mAP
Baseline-1	76.82	86.71	39.91
Baseline-2	80.82	89.57	49.68
Baseline-3-front	49.57	56.92	20.33
Baseline-3-rear	41.36	47.67	16.38
Baseline-3-side	29.79	33.35	10.15
Baseline-4-front-rear	60.25	68.58	26.43
Baseline-4-front-side	55.95	62.16	22.12
Baseline-4-rear-side	52.96	58.88	20.76
Multi-View	82.64	90.87	52.26

parameters and generates higher dimensional feature than the Multi-View and Multi-View-Binary. Therefore, Table I shows our multi-view feature learning strategy is more effective than the single-branch and multi-branch ensembling frameworks. We can also see that on *VehicleID*, Multi-View and Multi-View-Binary achieve similar performance. On the *VeRi-776* dataset, Multi-View outperforms the Multi-View-Binary by 0.51% in Rank-1 accuracy. This is because some vehicle images in *VeRi-776* show fuzzy viewpoints, *e.g.*, viewpoints between the front and side-view as shown in Fig. 2. The probabilistic labels are more accurate for those cases. Therefore, we choose the probabilistic vector instead of the one-hot vector as our viewpoint label.

In addition to the above comparisons with baseline models, we further train the Baseline-3 using one single view, and the Baseline-4 with two branches using two views, respectively. Table II summarizes their performance on *VeRi-776*. It is clear that, using portions of training data, *e.g.*, images with one or two specific views performs worse than the Baseline-1.

As shown in Fig. 3, our final feature is computed by firstly weighting feature of each branch, then fusing them with simple element-wise addition. Recent works [58], [59] on 3D object recognition fuses multi-view features with bilinear pooling. We have tested bilinear pooling with Baseline-2, which improves the Baseline-2 but still performs worse than our method. For instance, bilinear pooling boosts the Rank-1 accuracy of Baseline-2 from 64.81% to 65.46% on *VehicleID* dataset, still lower than our 66.06%. This shows that, bilinear pooling might be not suited for our task, where only one branch corresponding to the viewpoint of input image is

TABLE III

COMPARISON BETWEEN SOFTMAX LOSS AND TRIPLET LOSS ON *VehicleID* AND *VeRi-776*. IT IS CLEAR THAT, SOFTMAX LOSS ACHIEVES SLIGHTLY BETTER PERFORMANCE COMPARED TO TRIPLET LOSS. USING BOTH LOSSES TOGETHER COULD ACHIEVE THE BEST PERFORMANCE. S, T REPRESENT USING SOFTMAX LOSS AND USING TRIPLET LOSS

Dataset	VehicleID		VeRi-776		
Methods	r=1	r=5	r=1	r=5	mAP
Baseline-1+T	52.16	63.35	68.12	80.43	34.55
Baseline-1+S	57.74	69.06	73.78	84.17	37.26
Baseline-1+T+S	60.63	72.67	76.82	86.71	39.91
Baseline-2+T	58.37	68.64	70.51	82.69	43.78
Baseline-2+S	62.56	73.92	77.17	86.63	47.59
Baseline-2+T+S	64.81	75.84	80.82	89.57	49.68
Multi-View+T	61.17	71.93	78.46	86.56	47.04
Multi-View+S	64.34	73.67	81.86	88.20	49.56
Multi-View+T+S	66.06	76.13	82.64	90.87	52.26

TABLE IV

EFFECTS OF SHARED CONVOLUTIONAL LAYERS BY MULTIPLE BRANCHES ON *VehicleID* AND *VeRi-776*. THE PERFORMANCE ACHIEVED WITH DIFFERENT LAYER SHARING SETTINGS ARE ALSO ILLUSTRATED. "MULTI-VIEW-1" DENOTES THE MULTI-VIEW BRANCHES ARE INSERTED AFTER THE CONV-1 STAGE

Dataset	VehicleID		VeRi-776		
Methods	r=1	r=5	r=1	r=5	mAP
Multi-View-1	66.53	77.16	82.24	90.46	52.91
Multi-View-2	66.94	77.92	83.06	91.11	54.11
Multi-View-3	66.27	77.11	82.42	91.01	52.97
Multi-View-4	66.06	76.13	82.64	90.87	52.26
Multi-View-1+LAM	66.83	77.16	83.15	91.06	53.16
Multi-View-2+LAM	67.18	77.79	84.78	91.58	54.77
Multi-View-3+LAM	66.73	77.56	84.62	91.37	53.85
Multi-View-4+LAM	66.44	77.23	84.33	91.28	53.29
Multi-View-1+GAM	67.31	77.25	85.25	91.41	55.03
Multi-View-2+GAM	67.78	78.16	86.05	92.31	57.23
Multi-View-3+GAM	67.73	77.95	85.61	91.42	56.28
Multi-View-4+GAM	67.61	77.80	85.97	91.58	56.99
Multi-View-1+GLAM	68.03	78.37	86.09	91.36	55.66
Multi-View-2+GLAM	68.51	78.98	87.12	92.79	57.75
Multi-View-3+GLAM	68.46	78.39	86.78	92.51	56.66
Multi-View-4+GLAM	68.31	78.16	86.35	92.43	57.48

activated for feature extraction, and the other branches are depressed. This setting differs from the one in 3D object recognition [58], [59], where images with different viewpoints provide complementary cues and could be equally important in providing discriminative cues for classification.

2) *Effects of Classification Loss and Triplet Loss*: This experiment compares the performance of training using triplet loss, softmax loss, and their combination, respectively. Table III summarizes the results. It is clear that, joint supervision of triplet loss and softmax loss helps to train better deep Re-ID models, which is consistent with observations in many prior works [60]–[62]. In following experiments, we use both softmax loss and triplet loss to train the network.

3) *Effects of Shared Layers*: As shown in Fig. 3, different branches in MVAN share several convolutional layers. This experiment tests the effect of shared layers. Table IV compares

TABLE V

COMPARISONS OF MODEL SIZE AND COMPLEXITY. "PN" DENOTES THE NETWORK PARAMETER NUMBER (MILLION)

Dataset	VehicleID			VeRi-776	
Model	PN	r=1	r=5	r=1	mAP
VGG_CNN_M_1024 [12]	86.2	31.9	50.3	-	-
Siamese Visual [14]	24	36.83	57.97	41.12	29.48
VAMI [19]	87.3	47.34	70.29	77.03	50.13
MVAN(2-branches)	22.1	68.31	78.16	-	-
MVAN(3-branches)	29.2	-	-	86.35	57.48

effects of applying multi-view branch structure after conv1, conv2, conv3 and conv4 stages, respectively.

As shown in Table IV, applying multi-view branch structure after conv2 achieves the best performance, better than sharing conv1 and the independent training in Baseline-2. It indicates that, it is important to learn common visual cues by sharing bottom convolutional stages, as well as to learn viewpoint-specific cues with independent higher convolutional stages. In other words, high convolutional stages could be more task and viewpoint related. It is also interesting to observe that, sharing 1 convolutional stage corresponds to the largest number of network parameters. However, this large network does not achieve the best performance. This could be because the low-level feature learned by bottom layers are robust to view-point variances. Sharing bottom convolutional stages with multi-branches allows them to be better trained in multiple training tasks.

4) *Effects of Spatial Attention Module*: We adopt spatial attention module in every branch to improve the robustness and discriminative power of learned features. Table IV also shows effects of applying different variants of spatial attention module in our multi-view branches.

As shown by the results, adding either LAM or GAM improves the Re-ID performance. It is also clear that, GLAM, i.e., the combination of LAM and GAM achieves the best performance. On the *VehicleID* dataset, Multi-View-4+GLAM outperforms the baseline, i.e., Multi-View-4, by 2.25% in Rank-1 accuracy. On *VeRi-776*, Multi-View-4+GLAM achieves more substantial performance gain, i.e., 5.22% in mAP. Compared with GAM, the GLAM has the same parameter size, but performs better. This indicates that, the performance gains of GLAM are not come from increasing the parameter size, but from the complementary usage of the GAM and LAM. Therefore, we could conclude that, our proposed spatial attention model effectively boosts the Re-ID performance.

5) *Model Complexity*: This part compares the proposed MVAN with three vehicle Re-ID CNN architectures, i.e. VGG_CNN_M_1024 [49], ResNet50 [14], and VAMI [19], respectively, in model size and complexity. Note that, *VehicleID* and *VeRi-776* exhibit two (front and rear) and three viewpoints, respectively. We hence design a two branch MVAN for *VehicleID* and a three branch MVAN for *VeRi-776*, respectively.

Table V summarizes the comparison. It is clear that, our method with two branches is smaller than those three

TABLE VI

PERFORMANCE OF OUR APPROACH BASED ON RESNET50 BACKBONE.
WE APPLY MULTI-VIEW BRANCHES AFTER THE RES4 STAGE.
Multi-View DENOTES OUR PROPOSED ARCHITECTURE
WITHOUT THE ATTENTION MODULE

Dataset	VehicleID			VeRi-776		
Methods	r=1	r=5	mAP	r=1	r=5	mAP
ResNet50	66.25	78.51	70.01	87.18	94.04	62.97
Multi-View	68.11	81.38	73.75	89.92	95.52	66.99
MVAN	72.58	83.07	76.78	92.59	97.93	72.53

TABLE VII

COMPARISON WITH RECENT WORKS ON *VeRi-776* [21]. * MEANS
ADDITIONAL SPATIAL-TEMPORAL CUES ARE USED.
† DENOTES MODIFIED MODEL

Methods	r=1	r=5	mAP	Backbone
GoogLeNet [29]	52.12	66.79	17.81	GoogLeNet
FACT [21]	50.95	73.48	18.49	GoogLeNet
OIFE [16]	65.9	87.7	48.00	GoogLeNet†
Siamese Visual [14]	41.12	60.31	29.48	ResNet50
SCAN [18]	82.24	90.76	49.87	VGG16
RAM [17]	88.6	94.0	61.5	VGG_CNN_M
VAMI [19]	77.03	90.82	50.13	Self-design
MSVR [63]	88.56	-	49.30	MobileNets
FDA-Net [64]	84.27	92.43	55.49	Self-design
PRM(256 × 256) [35]	92.2	97.9	70.2	ResNet50
FACT+Plate-SNN+STR* [21]	61.44	78.78	27.77	GoogLeNet
OIFE+STR* [16]	68.3	89.7	51.42	GoogLeNet†
Siamese Visual+STR* [14]	54.23	74.97	40.26	ResNet50
Siamese+Path-LSTM* [14]	83.49	90.04	58.27	ResNet50
VAMI+STR* [19]	85.92	91.84	61.32	Self-design
PVSS* [47]	90.58	97.14	62.62	ResNet50
MVAN	86.35	92.43	57.48	VGG16
MVAN	92.59	97.93	72.53	ResNet50

competitors and achieves significantly better Re-ID accuracy. As shown by the results on *VeRi-776*, adding one extra branch does not bring significant parameter overheads, and effectively boosts the performance.

6) *Performance With Stronger Backbone*: Our method is compatible with different backbones. We hence further implement MVAN with the ResNet50 backbone. The branches in MVAN share convolutional layers before the res5 stage to seek a reasonable trade-off between feature discriminative power and model complexity.

We summarize the performance achieved with ResNet50 in Table VI. It is clear that, the two important components, *i.e.*, multi-view branch and spatial attention block, still boost the baseline performance. We could conclude that, our method works well with strong backbone and achieves better performance based on higher baseline.

D. Comparison With Recent Works

This part compares MVAN against recent works and summarizes the comparison on two datasets in Table VII and Table VIII, respectively.

1) *VeRi-776*: Comparisons on *VeRi-776* are summarized in Table VII, where GoogLeNet [29], FACT+Plate-SNN+STR [21] and Siamese-Visual [14] learn global features for vehicle Re-ID. OIFE [16], SCAN [18], VAMI [19] and FDA-Net [64] use attention modules to refine the

TABLE VIII

COMPARISONS WITH RECENT WORKS ON *VehicleID* [12].
† DENOTES MODIFIED MODEL

Methods	r=1	r=5	mAP	Backbone
GoogLeNet [29]	38.27	59.39	40.39	GoogLeNet
VGG + Triplet Loss [12]	31.9	50.3	37.3	VGG_CNN_M
VGG + CCL [12]	32.9	53.3	38.6	VGG_CNN_M
Mixed Diff + CCL [12]	38.2	61.6	45.5	VGG_CNN_M
VGG+C+T [60]	61.0	77.5	-	VGG_CNN_M
CLVR [65]	50.60	68.00	-	Inception-V3
OIFE [16]	67.0	82.9	-	GoogLeNet†
C2F [30]	51.4	72.2	53.0	GoogLeNet
VAMI [19]	47.34	70.29	-	Self-design
SCAN [18]	65.44	78.47	-	VGG16
MSVR [63]	63.02	73.05	-	MobileNets
FDA-Net [64]	55.53	74.65	61.84	Self-design
MVAN	68.31	78.16	73.03	VGG16
MVAN	72.58	83.07	76.78	ResNet50

feature maps in CNN. Siamese Visual+STR [14], Siamese-CNN+Path-LSTM [14], VAMI+STR [19] and PVSS [47] use spatial-temporal cues to further improve the Re-ID accuracy. MSVR [63] proposes a multi-scale method for various spatial resolutions of vehicle images.

In Table VII, our method consistently obtains better performance over other competitors on rank-1 and mAP. Note that, although some recent methods like VAMI+STR [19] and PVSS [47] use extra spatial-temporal cues, our method still outperforms them. VAMI [19] also considers viewpoint feature embedding by using adversarial training strategy. Compared to the VAMI, our learning scheme is easy to implement, and achieves better performance. PRM(256 × 256) [35] integrates part and global cues and obtains good performance. However, it needs additional bounding box detection network for part localization, which is not applicable for side view vehicle image. In addition, our performance is better than PRM(256 × 256) [35] as shown in Table VII.

2) *VehicleID*: Table VIII shows comparison with recent works on *VehicleID* dataset. Among those compared works, Mixed Diff+CCL [12] and VGG+C+T [60] propose an improved deep metric learning method for vehicle Re-ID task. C2F [30] proposes a novel coarse-to-fine ranking loss to learn a structured deep feature embedding.

As shown in Table VIII, MVAN outperforms all those competitors in both rank-1 and mAP. For example, compared with OIFE [16] which uses key point annotations and extra training data, our approach still achieves better performance. The recent FDA-Net [64] designs a feature distance adversary scheme to generate hard negative samples in feature space. It applies attention regularization to improve the discrimination capability on subtle region differences. Compared with FDA-Net, our approach achieves substantially better performance, *e.g.*, 68.31% vs. 55.53% in rank-1 accuracy. It is also clear that, with the stronger ResNet50 backbone, our method demonstrates further performance gains, *e.g.*, boosts the rank-1 accuracy to 72.58%.

The above experiments show the promising performance of MVAN. Fig. 6 demonstrates some sample vehicle Re-ID results on the *VeRi-776* dataset. The results of Baseline-1 are also illustrated in the first row for comparison. It is clear



Fig. 6. Visualization of sample vehicle Re-ID results on *VeRi-776* dataset. Top 10 returned images are shown. Red and green bounding boxes denote false positive and true positive, respectively. For example query, the first and second row shows Re-ID results by baseline and MVAN, respectively.

that, our method produces more reliable Re-ID results. It also could be observed that, the baseline tends to retrieve vehicles with similar viewpoint to the query. MVAN is more robust the viewpoint variances and effectively retrieves true positives with different viewpoints.

V. CONCLUSION

This paper proposes a Multi-View Attention Network (MVAN) for the challenging vehicle Re-ID task. We design a multi-view branch network to produce more discriminative viewpoint invariant feature. In addition, we design a two stream spatial attention block consisting of the global attention module and local attention module to alleviate cluttered backgrounds and discover subtle local differences among different vehicles. Extensive experiments show that, both the multi-view branch and global-local spatial attention module boosts the Re-ID performance on two widely used vehicle Re-ID datasets.

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